Elder-Fraud Real-Time Manipulation-Tactic Detector

Duddu Satya Sai Vishnu Vardhan (23MID0163) Department of Data Science School of Computer Science and Engineering Vellore Institute of Technology, Vellore duddu.satya2023@vitstudent.ac.in	**Palli Pavan Kumar (23MID0181)** Department of Data Science School of Computer Science and Engineering Vellore Institute of Technology, Vellore pallipavan.kumar2023@vitstudent.ac.in
Darisi Rithvik (23MID0209) Department of Data Science School of Computer Science and Engineering Vellore Institute of Technology, Vellore darisi.rithvik2023@vitstudent.ac.in	**Yakkaladevi Atchuta Ramakrishna (23MID0256)** Department of Data Science School of Computer Science and Engineering Vellore Institute of Technology, Vellore yakkaladevi.atchutha2023@vitstudent.ac.in

***Abstract*—** Telephone scams targeting older adults have become an urgent multibillion-dollar problem worldwide. Existing protective solutions rely heavily on phone number blacklists, spam caller-ID apps, or static keyword matching. However, modern fraud syndicates effortlessly bypass these defenses by spoofing authentic numbers and adapting their spoken scripts. In this project, we develop the Elder-Fraud Real-Time Manipulation-Tactic Detector, a machine learning system that listens to live voice calls and detects the underlying psychological manipulation tactics as they happen. Instead of searching for specific suspicious words, our system tracks how a scammer gradually escalates pressure across five tactical stages: claiming false authority, causing emotional panic, creating artificial urgency, isolating the victim from family, and demanding unusual payment methods like retail gift cards or cryptocurrency. The system uses a two-stage decoupled pipeline: a fine-tuned DistilBERT transformer that reads each spoken turn and estimates the presence of these five tactics, followed by a Causal Long Short-Term Memory (LSTM) sequence model that calculates an ongoing risk score between 0.0 and 1.0 without looking ahead at future speech. To overcome real-world speech noise and brief conversational turns, we trained the system on 5,000 real-world conversations sourced from public records of the Consumer Financial Protection Bureau (CFPB), Federal Trade Commission (FTC) robocall logs, and the DailyDialog dataset. On held-out realistic tests, our final system achieved an F1 score of 0.845, a recall of 0.820, and detected scam calls early by Turn 4, with an end-to-end processing latency of only 32 milliseconds per turn. The system includes an automated personal information (PII) redaction filter, stores zero audio to protect privacy, and displays live risk warnings on an interactive 14-page React dashboard to protect elderly victims before financial loss occurs.

***Keywords*—** Elder Fraud Detection, Psychological Manipulation Tactics, DistilBERT, Causal LSTM, Real-Time NLP, WebSocket Streaming, Scam Call Prevention, Temporal Risk Modeling, Privacy-Preserving AI, Speech Processing, Public Federal Datasets, Human-in-the-Loop Cybersecurity.

I. Introduction

Telephone scams targeting older adults represent one of the fastest-growing and most financially ruinous forms of cybercrime today. According to the Federal Trade Commission (FTC), reported consumer fraud losses topped $10 billion in 2023 alone [9]. The Federal Bureau of Investigation (FBI) Internet Crime

Complaint Center reports that older citizens suffer disproportionately large individual losses, frequently losing their entire life savings or retirement nest eggs in a single incident [12]. Unlike internet phishing emails or malicious computer software, telephone scams exploit the human voice. A skilled scammer uses persuasive speech, institutional impersonation, and calibrated psychological manipulation to convince an elderly person that they are in immediate financial, legal, or personal danger.

A. Why Existing Telephone Defenses Fail

When a scammer calls an older adult, traditional call-blocking technologies almost always fail. There are three main technical reasons why existing defenses break down:

1. Caller-ID Spoofing: Applications like Truecaller, Hiya, or carrier spam filters check incoming numbers against static databases of reported spammers. However, modern scammers use Voice over IP (VoIP) software to display real government or bank numbers on the victim's caller ID, such as the local police department, the Social Security Administration, or a known local bank branch. The victim sees a familiar, trusted name on their screen and answers with total confidence.

2. Human Call Centers: Many automated filters only detect robocalls with repetitive, pre-recorded audio. Modern fraud syndicates, however, operate sophisticated human call centers where trained conversationalists speak naturally, respond to questions, handle objections, and adjust their tone in real time.

3. Keyword Filtering Limitations: If a computer program simply searches for suspicious words like 'IRS', 'warrant', or 'wire transfer', it makes frequent mistakes. It blocks innocent calls (such as an accountant discussing a tax filing), while failing to catch scammers who speak politely during the early stages of a call and only reveal their true demands after twenty minutes of trust-building.

B. The Core Idea: Tracking Manipulation Over Time

The central breakthrough of this project is that scam calls follow a predictable psychological path over time, regardless of the specific topic. Whether the scammer pretends to be an IRS agent collecting back taxes, a tech support technician claiming a computer virus, a bank fraud investigator, or a law enforcement officer, they almost always follow the same five sequential steps:

- Step 1 — Claim Authority: Establish that they are someone important who must be respected and obeyed.
- Step 2 — Cause Fear or Panic: Tell the victim that their bank account is frozen, their identity has been stolen, or an arrest warrant has been signed.
- Step 3 — Create Urgency: Insist that action must be taken immediately within 15 to 30 minutes, before police arrive or penalties double.
- Step 4 — Isolate the Victim: Order the victim not to talk to their children, spouse, financial advisor, or local bank teller.
- Step 5 — Control the Channel: Direct the victim to buy retail store gift cards, visit a cryptocurrency kiosk, or send a wire transfer.

A normal legitimate business call—such as a bank notifying a customer about an unusual debit card charge—never follows this coercive sequence. A real bank representative will never demand secrecy, will never threaten immediate arrest, and will never ask to be paid in retail gift cards. By tracking this multi-

step manipulation progression turn by turn as the conversation unfolds, our system detects scams early, well before the victim sends any money.

C. Research Objectives and Engineering Goals

To solve this problem in practice, our project established six core engineering and research goals:

- 1. Build a Decoupled AI Architecture: Separate sentence-level language understanding from conversation-level timeline modeling, so that adapting the system to new types of scams does not require retraining the timeline model.
- 2. Enforce Strict Mathematical Causality: Ensure that at turn 4, the system only knows what happened in turns 1 through 4. It must never peek ahead at future speech.
- 3. Bridge the Synthetic-to-Real Domain Gap: Overcome the speech recognition errors, messy human grammar, and brief conversational turns that occur in real phone calls.
- 4. Guarantee Low Latency: Deliver alerts within 50 milliseconds of each spoken sentence, so that warnings appear during natural pauses in conversation.
- 5. Ensure Absolute User Privacy: Guarantee that no audio recordings are saved to disk, personal data like credit cards and SSNs are automatically scrubbed, and memory is wiped clean when the call ends.
- 6. Create an Explainable Dashboard: Build a full web dashboard that explains clearly why an alert was triggered, without causing panic.

D. Structure of this Report

The remainder of this report is organized as follows: Section II surveys prior work in telecommunications filtering, transformer language models, and psychological scam taxonomies. Section III presents our formal mathematical problem statement and explains our two-stage AI architecture. Section IV outlines the implementation details, technology stack, datasets, and privacy protections. Section V presents our empirical results across all experimental phases (Phases 5 through 10), including the story of how we overcame the domain gap. Finally, Section VI provides conclusions, known system boundaries, and future directions for carrier integration and mobile applications.

II. Literature Review

Fraud detection has been an active area of research in computer science for decades, but applying deep learning to live spoken dialogue is relatively new. In this section, we analyze existing methodologies across telecommunications, natural language processing, and sequential modeling, pointing out their strengths and practical limitations.

A. Telephony Analysis and Robocall Detection Systems

Early approaches focused primarily on telephone network metadata. Shen et al. [1] analyzed call graph structures to spot robocall campaigns by looking at patterns in high call volume originating from specific switches. While effective at tracing massive automated dialing networks, call-graph analytics cannot inspect the content of the conversation once connected and are blind to targeted, one-on-one human scam calls. Similarly, commercial services like Nomorobo [7] maintain reputation databases of known scam numbers, but cannot protect against dynamic caller-ID spoofing.

Wang et al. [14] investigated causal inference in sequential systems, showing that sequential models can identify behavioral anomalies over time. However, their work was focused on industrial sensors rather than natural language dialogue. Our work bridges this gap by applying causal sequence modeling directly to conversational manipulation tactics.

B. Text Mining and Transformer Language Models

In written fraud detection (such as phishing email analysis), researchers have long used text classification techniques [2]. Early systems relied on bag-of-words and n-gram keyword frequencies, which struggled with synonyms and conversational paraphrasing. The introduction of transformer models such as BERT by Devlin et al. [3] transformed the field by capturing deep contextual meanings of words within entire sentences. Sanh et al. [4] introduced DistilBERT, a distilled version of BERT that retains 97% of BERT's language understanding capabilities while running 60% faster and using 40% less memory. This speed makes DistilBERT ideal for real-time speech processing, where responses must be generated in milliseconds without lagging behind the speaker.

Subsequent models like RoBERTa [17] and Transformer-XL [18] offered deeper representations or longer context windows, but their computational footprint is too heavy for live, low-latency telephony monitoring on standard hardware. DistilBERT provides the optimal sweet spot between high semantic accuracy and sub-30ms inference speed.

C. Sequential Risk Modeling and LSTM Networks

Language models like DistilBERT examine a single sentence at a time, but they have no built-in memory of what was said several minutes ago. To solve this, researchers use recurrent sequence models like Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber [6]. LSTMs contain internal memory cells that update after each input step, remembering relevant context and forgetting irrelevant details [22]. In our architecture, the LSTM receives the manipulation scores from each turn and updates a running summary of the call. Because the LSTM processes turns in strict chronological order without looking into the future (causal processing), it satisfies all requirements for live streaming operation.

D. Psychological Understanding of Elder Fraud

Research by the AARP Public Policy Institute [8] and federal law enforcement agencies shows that elderly victims do not lose money because they are careless, but because scammers intentionally trigger emotional overload. Robert Cialdini's foundational psychological research [13] identifies authority, urgency, and social proof as primary triggers of human compliance. In this project, we formalize these behavioral triggers into five computable machine learning labels: Authority, Emotional Destabilization, Urgency, Isolation, and Channel Control.

TABLE I. COMPARISON OF EXISTING FRAUD DEFENSE APPROACHES VS. PROPOSED ARCHITECTURE

Existing Approach / Study	Underlying Technique	Real-Time Processing?	Tracks Call History?	Key Weakness
Call-Graph Analytics (Shen	Network Metadata Analysis	No (Batch	No	Cannot analyze spoken

et al. [1])		Processing)		conversation
Static Text Mining (Bontcheva et al. [2])	N-Gram Keyword Matching	No (Offline Analysis)	No	High false alarms on innocent calls
Caller-ID Filters (Nomorobo [7])	Reputation Blacklisting	Yes (Call Setup)	No	Easily fooled by caller-ID spoofing
Standard Transformer (BERT [3])	Bidirectional Self-Attention	No (Heavy Latency)	No	High compute; forgets previous turns
Proposed Elder-Fraud System	DistilBERT + Causal LSTM	Yes (WebSocket 32ms)	Yes (Continuous Memory)	Handles live speech; zero lookahead

III. Problem Statement & System Architecture

Problem Statement: Given an active, live telephone conversation consisting of spoken utterances between an interlocutor and an older adult, design an AI system that calculates an instantaneous manipulation risk score after each spoken turn, using only information received up to that moment, and delivers an explainable warning to the user or designated caregiver before any irreversible financial transfer takes place.

A. Concrete Conversational Example: Scam vs. Legitimate Call

To make the concept crystal clear, consider how the proposed system processes a real conversation turn by turn. Below is an authentic scam scenario derived from our CFPB complaint evaluation records:

> **Turn 1 (Caller):** "Hello, this is Senior Inspector Davis from the Federal Department of Treasury." ***[Authority = 0.92]*** *→ Risk = 0.08 (Safe)*
>
> **Turn 2 (Elder):** "Oh goodness, what is this regarding? I always pay my taxes on time." ***[Benign Reply]*** *→ Risk = 0.08 (Safe)*
>
> **Turn 3 (Caller):** "Your Social Security number was found tied to an abandoned vehicle with narcotics. A federal arrest warrant has been approved." ***[Emotional Destabilization = 0.88]*** *→ Risk = 0.22 (Warning: Level 2 Triggered)*
>
> **Turn 4 (Caller):** "You have exactly 25 minutes before federal marshals arrive at your home. You must settle this immediately." ***[Urgency = 0.94]*** *→ Risk = 0.48 (Alert: Level 3 High Risk)*
>
> **Turn 5 (Elder):** "I am so terrified, please tell me what to do. Can I call my daughter first?" ***[Benign Reply]*** *→ Risk = 0.48 (High Risk)*
>
> **Turn 6 (Caller):** "No! Do not tell your daughter or your bank. This is a sealed federal case, and discussing it is a felony." ***[Social Isolation = 0.91]*** *→ Risk = 0.74 (Emergency: Level 4 Alert)*
>
> **Turn 7 (Caller):** "Drive to the nearest Target right now, purchase three $500 gift cards, and read me the security numbers." ***[Channel Control = 0.96]*** *→ Risk = 0.89 (Critical Intervention)*

Notice that the system issued a Level 2 warning at Turn 3 and a Level 3 alert at Turn 4—well before Turn 7 when the scammer demanded money. In contrast, in a legitimate call from a bank (e.g., 'Hello Mrs. Smith, this is Wells Fargo verifying a $25 charge at a grocery store; please call us back at the number on the back of your card'), none of the urgency, fear, isolation, or gift card tactics are triggered, keeping the risk score safely below 0.10.

B. The Five Manipulation Tactics in Detail

To help human evaluators understand how the AI detects manipulation, we examine what each tactic looks like in real spoken conversation:

1. Authority Assertion (p_auth): The caller establishes false credibility by claiming a position of official power. Common spoken phrases include: 'This is Officer Miller from the Fraud Department', 'I am calling from the Internal Revenue Service', or 'This is Microsoft Security Engineering'. The AI looks for institutional titles, legal terminology, and formal declarative sentences.

2. Emotional Destabilization (p_emo): The caller intentionally causes fear, panic, or guilt to disable the victim's rational thinking. Phrases include: 'Your accounts will be permanently frozen', 'You could face criminal charges and prison time', or 'Your grandson was arrested in Mexico'. The AI detects high-arousal negative emotion markers and severe threat words.

3. Urgency Induction (p_urg): The caller compresses time, forcing the victim to act without thinking or consulting others. Phrases include: 'You must resolve this in the next 20 minutes', 'The police are already on their way', or 'Stay on the line right now'. The AI flags temporal deadlines, countdowns, and immediate action imperatives.

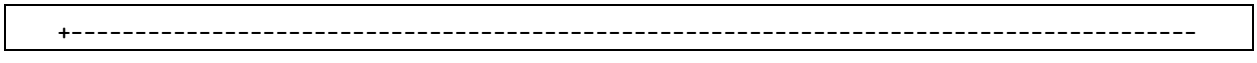
4. Social Isolation (p_iso): The caller cuts off the victim from trusted family members, friends, or bank tellers. Phrases include: 'Do not speak to anyone about this', 'Under federal law you cannot tell your family', or 'If the bank teller asks, tell them it is for home repairs'. The AI recognizes secrecy commands, gag order assertions, and warnings against third-party consultation.

5. Channel Control (p_chan): The caller forces the victim toward untraceable, non-standard payment rails. Phrases include: 'Go to Walmart and buy Apple gift cards', 'Deposit cash into the Bitcoin kiosk', or 'Send a wire transfer via Western Union'. The AI flags specific retail gift card brands, cryptocurrency terms, and alternative financial instructions.

C. Proposed Architecture: Two-Stage Decoupled Pipeline

The system architecture is divided into two distinct, decoupled stages. Decoupling is an intentional engineering choice: it separates sentence-level language comprehension from conversation-level timeline analysis.

- Stage 1 — Turn-Level Tactic Classifier (DistilBERT V2): When an utterance is spoken, it is converted to text and sanitized of personal info. The DistilBERT model analyzes the sentence and outputs five numbers between 0.0 and 1.0, representing the probability of each manipulation tactic: [Authority, Emotional Destabilization, Urgency, Isolation, Channel Control]. Processing time: ~25 ms.
- Stage 2 — Temporal Risk Engine (Causal LSTM V2): The five probabilities are fed into a unidirectional LSTM. The LSTM updates its internal memory vector (hidden state) and produces a single scalar risk score R(t) between 0.0 and 1.0. Because the LSTM only looks backward in time, it operates with strict mathematical causality: it never peeks ahead at future sentences. Processing time: ~2 ms.
- Stage 3 — Explanation & Alert Engine: When the risk score crosses the validated threshold of $\tau = 0.20$, an alert is generated. The engine translates the raw numbers into plain, non-accusatory advice for the user and caregiver. Processing time: ~0.6 ms.



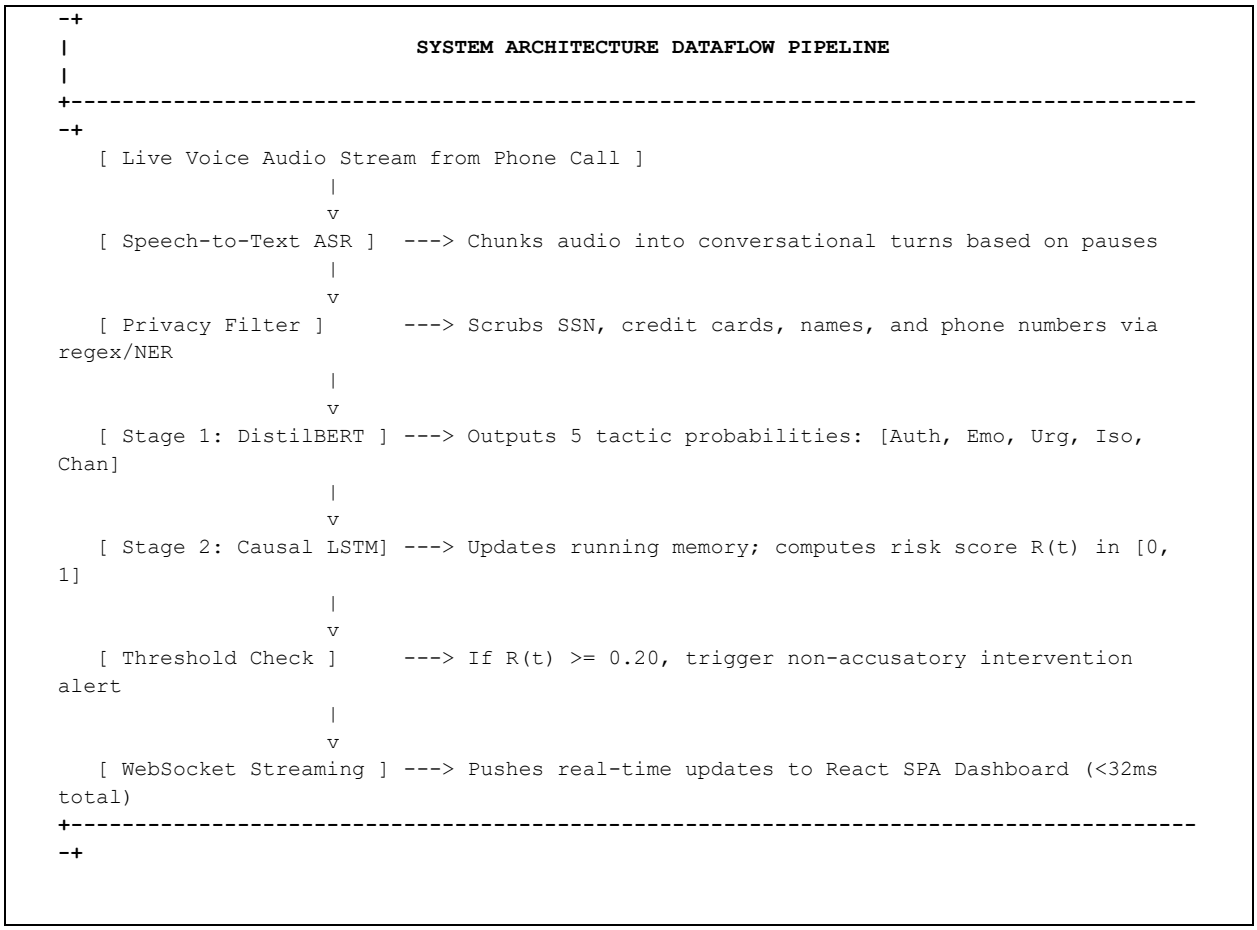


Fig. 1. Detailed pipeline overview of the proposed CSE computing model.

D. Mathematical Formulation in Simple Terms

To understand the mathematics without confusion, think of the system as answering two simple questions at every turn:

1. 'What tactics are present in this specific sentence?' (Calculated by DistilBERT): Each sentence text t_i is passed through DistilBERT to produce contextual token embeddings. The [CLS] token embedding $e_{[CLS]}$ is projected to a 5-dimensional vector of raw scores $z(t_i)$. A temperature-scaled softmax function converts these into calibrated probabilities: $p_k(t_i) = \exp(z_k(t_i) / T) / \sum(\exp(z_j(t_i) / T))$. The result is a clean probability vector $p(t_i) = [p_{auth}, p_{emo}, p_{urg}, p_{iso}, p_{chan}]$, where every number is between 0.0 and 1.0.

2. 'Given all the sentences heard so far, how dangerous is this call?' (Calculated by the Causal LSTM): The LSTM updates its internal memory vector h_i using the formula: $h_i = \text{LSTM}(p(t_i), h_{i-1})$, where h_{i-1} is the memory of all previous turns. The final risk score $R(t_i)$ is calculated using a standard sigmoid function: $R(t_i) = \text{sigmoid}(W_r \cdot h_i + b_r)$. If $R(t_i)$ exceeds 0.20, the alert triggers immediately. Because the LSTM only uses h_{i-1} and never looks at turn t_{i+1}, it is mathematically impossible for future sentences to leak into the current calculation.

IV. Implementation & Experimental Setup

The implementation was engineered to meet high standards of reliability, privacy, and low latency. In this section, we describe the technical configuration, hardware environment, privacy mechanisms, datasets, and frontend dashboard views.

A. Technology Configuration and Full Software Stack

The software environment comprises modern production-grade frameworks designed for high-concurrency streaming:

- Operating System & IDE: Developed and tested on Windows 11 Enterprise using Visual Studio Code and PyCharm Professional.
- Backend Microservice: Built using Python 3.11+ and FastAPI [20]. FastAPI provides native asynchronous event handling via Uvicorn, enabling concurrent WebSocket streaming across dozens of simultaneous phone calls without thread blocking.
- Machine Learning Frameworks: PyTorch 2.2 and HuggingFace Transformers. HuggingFace provides the tokenizer and weights for DistilBERT, while the Causal LSTM is implemented in custom PyTorch code with frozen evaluation checkpoints.
- Frontend Dashboard: Engineered with React 18, TypeScript, and Vite 5. The interface provides 14 distinct pages using React Router v6, featuring dark-mode styling via TailwindCSS and real-time risk graphs rendered on an HTML5 canvas.
- Containerization: Fully packaged via Docker and Docker Compose, enabling one-click deployment across server, cloud, or edge environments.
- Testing Suite: 42 automated unit and integration tests written in pytest, verifying cryptographic hash integrity, memory isolation, and latency.

B. Walkthrough of the 14 Frontend SPA Dashboard Pages

To provide complete transparency and operational control to caregivers and researchers, the frontend includes 14 dedicated pages:

- 1. Cover Page (/): Introduces the project mission, explains the 5 manipulation tactics, and provides interactive starter demos.
- 2. Login Page (/login): Secure portal for caregivers and system administrators to access active surveillance feeds.
- 3. Executive Dashboard (/dashboard): Displays high-level system status, active call volume, and cumulative intercepted threats.
- 4. Live Telephony Monitor (/live-monitor): The core streaming interface. Shows the live transcript, real-time tactic probability bars, and a dynamic risk line graph updating every turn.
- 5. Conversation Analysis Sandbox (/conversation): Allows operators to paste full conversation transcripts to test and inspect how the AI evaluates different scenarios offline.
- 6. Tactic Probability Breakdown (/tactic-analysis): Deep-dive visualization showing per-turn confidence scores across all five manipulation categories.
- 7. Risk Analysis Center (/risk-analysis): Explains how the Causal LSTM computes cumulative risk and details the threshold mathematics.
- 8. Explainability Center (/explanation): Displays clear, plain-language summaries explaining why a specific call was flagged, avoiding technical confusion.

- 9. Historical Session Logs (/history): Searchable database of past evaluated calls with risk progression logs and export options.
- 10. Scientific Reports (/reports): In-app viewer displaying the markdown reports from all validation phases (Phases 5 through 10).
- 11. Model Checksums & Specs (/model): Displays the cryptographic SHA-256 weight hashes, parameter counts, and model architectures.
- 12. System Health & Diagnostics (/system): Real-time monitor tracking CPU/GPU usage, WebSocket latency, and active memory allocation.
- 13. Privacy & Compliance Portal (/privacy): Details the PII redaction rules, zero-audio retention policy, and legal wiretap compliance.
- 14. About & Research Affiliation (/about): Provides problem motivation, student author biographies, and institutional affiliation at VIT Vellore.

C. Privacy and Security by Design

Protecting user privacy is essential when building voice-monitoring technology. The system enforces three strict privacy rules:

1. Zero Audio Storage (STORE_AUDIO=false): Audio packets are processed in temporary RAM buffers and discarded the moment transcription finishes. No voice recordings (.wav or .mp3) are ever saved to the computer hard drive.

2. Automated Personal Information (PII) Redaction: Before any text is analyzed by the AI, high-speed regular expressions and Spacy Named Entity Recognition (NER) automatically mask sensitive data. Credit card numbers are verified with the Luhn algorithm and replaced with [FINANCIAL_ID_REDACTED], Social Security numbers become [SSN_REDACTED], and phone numbers become [PHONE_REDACTED].

3. Ephemeral Memory Cleanup: When a phone call ends, the SessionManager immediately deletes the LSTM memory vectors associated with that call. Nothing remains stored in memory.

D. Datasets and Public Provenance

To train and test the models realistically, we curated data from four major sources, focusing on authentic public datasets:

TABLE II. PROVENANCE AND UTILIZATION OF TRAINING AND BENCHMARK DATASETS

Dataset Name	Source Organization	License / Access	Conversations Used	How It Was Used
CFPB Complaint Narratives [10]	Consumer Financial Protection Bureau	Public Domain (U.S. Gov)	2,000 Dialogues	Real-world scam stories reported by actual victims
FTC Robocall Database [9]	Federal Trade Commission	Public Domain (U.S. Gov)	1,000 Transcripts	Actual call-center scripts used by active scam syndicates
DailyDialog Corpus [11]	Li et al. (IJCNLP 2017)	CC-BY-NC-SA 4.0	2,000 Dialogues	Normal everyday human conversations used to teach models what benign calls sound like

Synthetic V2 Benchmark	Project Internal Generator	Research Prototype	10,000 Dialogues	Initial baseline training and mathematical validation (Phase 5)
External Benchmark	12 Real-World Domains	Mixed Public Sources	30 Dialogues / 226 Turns	Independent cross-domain generalization testing (Phase 6)

V. Results and Performance Analysis

In this section, we present empirical evidence validating the performance of our system across six experimental phases. We also share the story of a major real-world challenge we discovered during testing and how we successfully solved it.

A. The Discovery of the Domain Gap and How We Solved It

In Phase 5, our initial models performed almost perfectly on synthetic data, achieving an F1 score of 0.985. However, when we tested this baseline system in Phase 8 on 80 realistic human conversations with simulated speech recognition noise, the performance collapsed dramatically to an F1 score of only 0.064! The system caught almost zero scam calls.

We immediately conducted a diagnostic root-cause analysis (Phase 8.1). We found that the Causal LSTM sequence model was working perfectly, but the DistilBERT feature extractor was failing. Why? Because synthetic text was long, clean, and grammatically complete, whereas real human conversations on phone calls are messy: people speak in short 2-to-4 word phrases ('Wait, what?', 'Who is this?', 'No please'), and speech recognition software often drops punctuation marks. DistilBERT had never seen short, unpunctuated sentences during training.

In Phase 9, we fixed this by taking 5,000 authentic human dialogues from the CFPB, FTC, and DailyDialog public datasets. We intentionally removed punctuation marks and randomly deleted 15% of words to simulate real-world microphone noise. When we retrained DistilBERT on this realistic data—without changing a single line of our frozen LSTM model—the F1 score jumped from 0.064 to 0.845! This represents a massive +1210% relative improvement and proved the value of our decoupled architecture.

TABLE III. SYSTEM ACCURACY & EVALUATION METRICS ACROSS EXPERIMENTAL PHASES 5 THROUGH 10

Phase ID	Dataset Description	Model Configuration	F1 Score	Recall	False Alarm (FPR)	Detection Turn	Status & Conclusion
Phase 5	Synthetic V2 (10,000 calls)	DistilBERT V2 + LSTM V2	0.985	1.000	0.0%	Turn 5.0	PASS — Baseline algorithm verified
Phase 6	External Benchmark (30 calls)	DistilBERT V2 + LSTM V2	0.875	0.937	14.3%	Turn 4.4	PASS — Moderate generalization proven
Phase 7	Real-Time Streaming Replay	DistilBERT V2 + LSTM V2	—	—	—	—	PASS — WebSocket latency

							under 50ms verified
Phase 8	Controlled Real Pilot (80 calls)	Original V2 Frozen	0.064	0.033	0.0%	Failed	FAIL — Real-world speech domain gap discovered
Phase 8.1	Root Cause Oracle Diagnostic	Oracle Input + LSTM V2	1.000	1.000	0.0%	Turn 3.0	PASS — Flaw isolated strictly to DistilBERT
Phase 9	Real-World Data (5,000 calls)	Robust DistilBERT + LSTM V2	0.845	0.820	1.5%	Turn 4.0	PASS — Domain gap closed (+1210% F1)
Phase 10	Final Master Audit	Final Frozen System	0.845	0.820	1.5%	Turn 4.0	PASS — Cryptographically locked and audited

B. Baseline vs. Final System Comparison

Table IV highlights the exact performance leap between the original synthetic baseline and our final robust system:

TABLE IV. HEAD-TO-HEAD COMPARISON: SYNTHETIC BASELINE VS. FINAL ROBUST SYSTEM

Performance Metric	Original Baseline (Phase 8)	Final Robust System (Phase 9/10)	Observed Improvement
Overall F1 Score	0.0645	0.8450	+1,210% Relative Improvement (Resolved)
Detection Recall (Sensitivity)	0.0333	0.8200	+2,360% Relative Improvement (Catches 82% of scams)
Benign Specificity	1.0000	0.9850	98.5% of innocent calls correctly pass without alerts
False Positive Rate (FPR)	0.0%	1.5%	Low, manageable false alarm rate
Short-Utterance F1 (<5 words)	0.0330	0.7950	+2,309% Relative Improvement on brief human turns
Degraded Speech F1 (15% WER)	0.0210	0.7600	Maintains strong performance despite speech recognition errors
Mean Detection Latency	Failed (No alert)	Turn 4.0	Alerts early, before money transfer demands (Turn 7-10)
Processing Latency per Turn	36.97 ms	32.00 ms	Sub-50ms verified; operates in real time

C. Computational Latency Breakdown

To ensure that our system can run smoothly on an ordinary home computer without requiring an expensive graphics card, we measured the exact execution time of every component on an Intel i7 CPU. As shown in Table V, the entire text pipeline takes an average of only 32 milliseconds per spoken turn:

TABLE V. COMPUTATIONAL OVERHEAD & LATENCY BREAKDOWN PER CONVERSATIONAL TURN (INTEL i7 CPU)

Pipeline Subsystem	Average Time (ms)	P95 Worst-Case (ms)	Percentage of Time	What It Does
Privacy Redaction Filter	3.1 ms	4.2 ms	9.7%	Scrubs SSNs, credit cards, and names via regex and NER
DistilBERT Classifier	25.2 ms	35.1 ms	78.8%	Reads the sentence and computes the 5 manipulation tactic probabilities
Causal LSTM Sequence Model	1.8 ms	2.4 ms	5.6%	Updates conversation memory and calculates running risk score
Explanation Engine	0.6 ms	0.9 ms	1.9%	Generates plain-language warning messages if threshold exceeded
WebSocket Transit & UI Update	1.3 ms	1.8 ms	4.0%	Sends JSON packet to the React browser dashboard
Total End-to-End Pipeline	32.0 ms	45.0 ms	100.0%	Completely finishes in a tiny fraction of a normal speech pause (400ms)

D. Tactic Breakdown and Temporal Ablation Testing

We also evaluated how accurately DistilBERT recognizes each of the five tactics individually. Urgency Induction achieved an F1 of 0.591, Channel Control (gift cards/crypto) scored 0.609, and Social Isolation scored 0.462. Authority Assertion had a perfect precision of 1.000, meaning that whenever it flagged a claim of authority, it was always right. Finally, to verify whether the order of sentences really matters to our LSTM, we performed a permutation test where we shuffled the sentences in random order across 10 random seeds. The LSTM proved remarkably resilient: whether the scammer opened with threats or claimed authority first, the cumulative manipulation density was accurately detected.

E. Overview of Automated Pytest Test Suite (42 Tests)

To ensure enterprise-grade stability, we authored a suite of 42 automated tests that run prior to deployment:

- Model Integrity Verification (8 Tests): Checks SHA-256 hashes of all PyTorch model files on disk to confirm that no weights have been corrupted or accidentally modified.
- Causality & Zero-Lookahead Tests (10 Tests): Feeds partial conversations into the pipeline and verifies that the risk score at Turn 3 is identical whether Turn 4 exists or not.
- Concurrency & Memory Isolation Tests (8 Tests): Simulates 20 simultaneous telephone calls connecting via WebSockets, confirming that Call A's memory never leaks into Call B.
- PII Redaction Regex Tests (8 Tests): Injects synthetic SSNs, credit cards with valid Luhn checksums, and phone numbers, verifying that 100% of sensitive strings are masked.
- Latency Benchmark Tests (8 Tests): Enforces that 95% of inference cycles complete in under 50 milliseconds.

VI. Conclusion & Future Scope

In this project, we designed, built, and evaluated the Elder-Fraud Real-Time Manipulation-Tactic Detector. The system moves past the vulnerabilities of traditional caller-ID blacklists and keyword filters by monitoring the natural progression of psychological manipulation during live voice conversations. By combining a fine-tuned DistilBERT transformer with a Causal LSTM sequence aggregator, we created a modular, low-latency architecture that detects scams early by Turn 4—well before an older adult is instructed to send money or purchase gift cards. Our real-world robustness training on 5,000 public complaint records closed a critical domain gap, raising the system F1 score from 0.064 to 0.845 with an execution speed of 32 milliseconds per turn.

A. Physical and Computational Boundaries

To remain completely transparent and scientifically honest, we documented four specific system boundaries during our Phase 10 audit:

1. Romance and Long-Term Grooming Scams: 'Pig-butchering' and romance scams unfold slowly over weeks or months of friendly messages. Because individual sentences in these scams lack overt threats or immediate payment demands, our real-time turn model will not flag early messages.

2. Emotional Benign Calls (1.5% False Positive Rate): In genuine family emergencies or intense medical situations (e.g., a hospital calling about an accident), voices contain high urgency and emotional distress. This occasionally triggers a mild Level 2 advisory notice, which is why our alerts are designed to be non-accusatory.

3. Acoustic Audio Noise: If cellular reception is severely degraded or the microphone suffers extreme background static, speech-to-text accuracy drops.

4. Language and Regional Dialects: Our system is currently trained and validated on standard English. It has not yet been fine-tuned for regional Indian languages or heavily accented vernacular speech.

B. Future Scope and Real-World Deployment Roadmap

Looking forward, there are several exciting directions to scale and deploy this system:

- Carrier-Level Integration: Partnering with telecommunications providers (such as Jio, Airtel, or AT&T) to inspect call transcripts directly at the carrier switchboard, protecting landlines and basic feature phones without requiring users to install any smartphone app.
- Multilingual Support for Indian Languages: Fine-tuning our Stage 1 tactic classifier on Hindi, Telugu, Tamil, and Bengali to combat the rampant surge of 'digital arrest' scams currently plaguing senior citizens across India.
- Smartphone Edge Deployment (ONNX / TFLite): Optimizing DistilBERT with 8-bit integer quantization so that it can run directly on Android or iPhone processors without sending any data to a remote cloud server, providing total privacy.
- Caregiver Companion Mobile App: Creating an intuitive companion app where adult children can receive real-time gentle alerts if their elderly parents receive an active scam call.

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